ModelIt!

Student Activity Pack

Your Phone Speaks in 1s and 0s

How Digital Signals Carry Music, Messages, and Memes

Grade 7 | NGSS MS-PS4-3 | Systems & Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is the difference between an analog signal and a digital signal? Give an example of each.

2 Why might a digital copy of a song sound different from the original live performance?

3 What does 'compression' mean when talking about digital files like MP3s?

4 Why do some music files take up more storage space than others?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sampling Rate | Compression Level | Signal Accuracy | File Size

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Studio Quality

Sampling Rate: HIGH | Compression Level: LOW

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Phone Streaming

Sampling Rate: MEDIUM | Compression Level: MEDIUM

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Extreme Compression

Sampling Rate: MEDIUM | Compression Level: HIGH

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Bit Depth — The number of possible values for each sample. Higher bit depth captures more subtle volume differences, like the difference between a whisper and a shout.

Transmission Speed — How quickly data can be sent from server to device. Faster connections allow higher quality streaming without buffering.

Error Correction — Digital systems can detect and fix errors during transmission. This is why digital signals are more reliable than analog — damaged bits can be reconstructed.

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why does your music sound perfect every time you play it — but your grandparents' vinyl records got scratchy? Use evidence from your simulation to support your answers.

1 Why are digital signals considered more reliable than analog signals for storing and transmitting information?

2 What trade-offs do engineers make when choosing sampling rate and compression settings?

3 How does the concept of sampling rate connect to the quality of digital music?

4 Why might a musician prefer vinyl (analog) even though digital is technically more reliable?

5 How has the shift from analog to digital changed how we consume music, movies, and information?

8 STEM Challenge: The Streaming Service Challenge

ModelIt!

YOUR ENGINEERING MISSION

Design a music streaming service that delivers the best possible sound quality while keeping data usage low enough for mobile users.

THE SCENARIO

Your startup is launching a music streaming app. Users on Wi-Fi want studio quality, but mobile users have limited data plans. You need to design a system that automatically adjusts quality based on connection type — without making the music sound terrible.

Engineering Design Thinking

1. WHAT CRITERIA DEFINE 'GOOD ENOUGH' QUALITY FOR MOBILE STREAMING?
2. HOW WILL YOUR SYSTEM DETECT AND ADAPT TO DIFFERENT CONNECTION SPEEDS?
3. WHAT HAPPENS WHEN A USER SWITCHES FROM WI-FI TO MOBILE MID-SONG?
4. HOW CAN YOU TEST WHETHER USERS CAN ACTUALLY HEAR THE DIFFERENCE BETWEEN QUALITY LEVELS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Audio Engineers use signal processing to record, mix, and master music. They understand sampling rates, compression, and digital audio workstations (DAWs). They work in recording studios, live concerts, film production, and video game audio — combining physics, math, and creativity.